

Lavish Devpura

B.Tech Computer Science Engineer

63 Jogi Talab, opp Step by Step School
Udaipur, Rajasthan

+91 8829968373

✉ lavishdevpura6@gmail.com

Profile Highlights

- Built an AI assistant using Retrieval-Augmented Generation with hybrid retrieval and tool orchestration.
- Experience deploying local LLMs including Mistral 7B using Ollama.
- Understanding of deep learning architectures such as Recurrent Neural Networks and Transformers.
- Experience building backend AI services using FastAPI.

Technical Skills

Programming Python (basic)

Machine Learning & AI Recurrent Neural Network, Transformer (deep learning architecture), Retrieval-Augmented Generation, Attention mechanisms, Semantic embeddings

Data Retrieval & RAG ChromaDB, BM25 hybrid search, Document chunking, Context retrieval

Backend & APIs FastAPI, Uvicorn, REST API

Tools Git

Work Experience

June **Data Science Intern, Internshala**

2025–July ○ Worked on real-world projects involving Big Data analysis using PySpark and Dask.

- 2025 ○ Developed predictive machine learning models and interactive dashboards with Power BI and Tableau.
○ Gained hands-on experience in applying data science concepts to solve business problems.

June **Data Science Trainee, Upflairs**

2024–July ○ Learnt about Machine Learning, Python Libraries, and Data visualization.

- 2024 ○ Worked with data to understand the significance of data science and predictive data analysis using machine learning.

Projects

Current **Hybrid RAG AI Assistant**, *Local LLM & Tool Automation*

- Built a tool-orchestrated AI assistant powered by a locally deployed Mistral 7B using Ollama for private, offline LLM inference.
- Designed a RAG pipeline with hybrid retrieval combining ChromaDB vector search and BM25 lexical ranking to improve context relevance.
- Developed backend APIs using FastAPI served with Uvicorn, exposing REST endpoints for chatbot interaction and tool execution.
- Implemented semantic intent routing to dynamically trigger multiple tools including news, sports, weather, and internet search APIs.
- Built system automation capabilities for executing GitHub commands and local file-system operations (macOS).
- Added voice interaction APIs for speech input and response generation, enabling voice-controlled chatbot interaction.
- Implemented reranking and context filtering to select relevant document chunks, improving response accuracy and efficiency.

Education

2022–2026 **B.Tech. Computer Science & Engineering**, *Amity University Rajasthan, Jaipur*,
GPA: 6.91

2022 **12th Grade**, *St Mary's Convent Sr Sec School*, 73%

2020 **10th Grade**, *St Mary's Convent Sr Sec School*, 77%